

# Lalit Maurya

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## PROFILE

Computer Science undergraduate with experience in full-stack development and AI/ML, skilled in Rust, Python, and modern web technologies. Passionate about building scalable, real-world solutions.

## EDUCATION

### Shiv Nadar Institution of Eminence

Bachelor of Technology in Computer Science and Engineering (CGPA 8.39)

Delhi NCR, India

Aug. 2023 – Expected May 2027

## EXPERIENCE

### Teaching Assistant - Computer Organisation and Architecture

On Campus Job

Aug 2025 – Nov 2025

Shiv Nadar Institution of Eminence

### Team Lead, AI/ML Team

Google Developer Group

March 2025 – Present

Shiv Nadar Institution of Eminence

- Created the official AI/ML team under GDG-SNIE, initiating and managing bi-weekly meetings to ensure project alignment and timely progress
- Directing a 13-member team of varying skill levels and diverse interests in real-world AI/ML projects.
- Spearheading the development of a campus-wide technical newsletter to showcase AI/ML projects and foster a culture of innovation and knowledge-sharing.

### Hackathon Director, @ping

Google Developer Group

April 2024 – Feb 2025

Shiv Nadar Institution of Eminence

- Created and led @ping, a campus-wide mini-hackathon series fostering an active development culture among fellow students, amassing 100+ participants and learners over a year
- Managed a 12-member team to develop a full-stack submissions platform using Supabase, and SvelteKit
- Designed and implemented a streamlined process for project submissions, improving user experience

## PROJECTS

### oxigrad-autograd | Rust, Python, PyO3, Maturin

Jul 2025

- Developed a scalar-valued auto-differentiation engine in Rust for memory-safety and performance
- Implemented Python bindings using PyO3 and Maturin for seamless integration across languages
- Implemented forward and backward passes, supporting automatic differentiation and gradient computation through dynamic computation graphs
- Integrated common activation functions (Sigmoid, ReLU, Softmax) and loss functions (MSE, Cross Entropy)
- Published to PyPI with support for Python 3.9+, enabling simple installation via `pip` or `uv`

### Skin Detection Without Color Information | OpenCV, Scikit-learn, NumPy

Feb 2025 - May 2025

- Implemented grayscale skin detection using facial priors and texture features from a WACV 2017 paper
- Processed and curated a HELEN-derived dataset with ground-truth skin segmentation masks
- Extracted LBP, grayscale stats, lacunarity, and landmark-based features
- Developed a region growing algorithm seeded by face-based intensity priors

### QueenSweep | Rust, Web Assembly, Typscript, Chromium Extension

Oct 2025 - Dec 2025

- Built a constraint satisfaction solver for LinkedIn Queens with aggressive pruning via 1-step lookahead and pluggable heuristics, solving 90% of puzzles in <10ms (p90: 9.55ms, 250+ states/ms)
- Designed zero-copy state representation using `Rc<[T]>` for immutable data sharing, reducing memory overhead by eliminating redundant clones during backtracking search
- Implemented WebAssembly bindings to compile Rust solver to browser-compatible binary, achieving near-native performance in JavaScript runtime
- Developed Chrome extension with DOM mutation observers for SPA navigation detection and synthetic event dispatch for automated solution application on [queensgame.vercel.app](https://queensgame.vercel.app)

### UniSync | Python, Selenium, Google Calendar API

Jan 2026

- Developed a python-based tool to streamline university timetable management by scraping data from the ERP system and syncing it with Google Calendar
- Leveraged Python, Selenium, and Google Calendar API to automate data extraction and integration, ensuring a seamless user experience.

## TECHNICAL SKILLS

**Languages:** Rust, JavaScript, TypeScript, Python, C, C++, SQL, HTML, CSS

**Frameworks & Libraries:** React, Next.js, SvelteKit, Flask, FastAPI, PyTorch, Poem-OpenAPI

**Tools & Platforms:** Git, GitHub, Supabase, Docker, Google Cloud Platform (GCP), Linux, Helix, Neovim, uv

**Databases & ORMS:** MySQL, PostgreSQL, MongoDB, Prisma, SQLx

**Technologies & Protocols:** Node.js, REST APIs, gRPC, WebSockets, Selenium